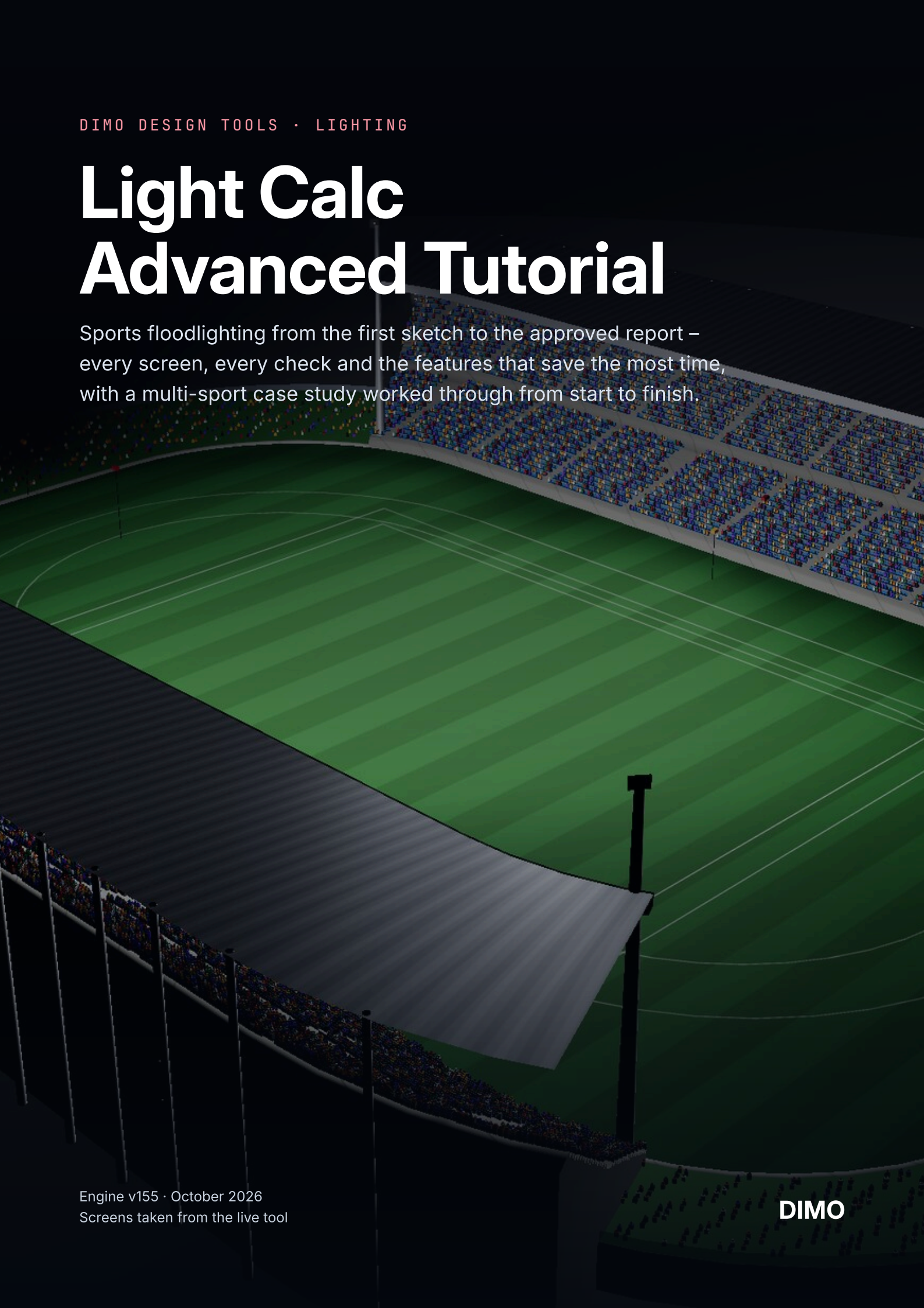
DIMO DESIGN TOOLS · LIGHTING

Light Calc Advanced Tutorial

Sports floodlighting from the first sketch to the approved report – every screen, every check and the features that save the most time, with a multi-sport case study worked through from start to finish.



Engine v155 · October 2026
Screens taken from the live tool

DIMO

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ABOUT THIS TUTORIAL

Every screen in it was taken from Light Calc engine v155, working on real designs: a football ground at EN 12193 Class I in Part B and a fictional multi-sport stadium in Part D. Results are the tool's own; figures such as luminaire colour data and the houses round the case-study ground are invented for the example and marked as such. The demo floodlights are generic – use the maker's IES or LDT file for a real design.

PART A

Getting around

What Light Calc does, how the screen is laid out, and the three habits that keep work fast: calculate on demand, undo freely, let the tool keep the design.

CHAPTER 1

What Light Calc does

Light Calc designs and checks outdoor sports floodlighting – masts or roof-mounted – against the standard you choose, point by point from the luminaires' photometric files, and produces the report, aiming schedule and exports a contractor and an approving engineer need.

1
Sport & standard
Cricket, football, rugby, athletics, tennis, pickleball, volleyball, or several sports on one site

2
Ground
Standard dimensions, OpenStreetMap, or your own drawing traced to scale

3
Luminaires
IES / LDT files; several to compare or mix beams

4
Masts or roof
Placed by you or by the tool, within the mast rules; or a roof catwalk

5
Automatic design
Heights, numbers, aiming and optimising until every target passes

6
Scenes & DMX
Match, training, televised, security – groups, levels, maximum demand

7
Checks
TV cameras, glare, neighbours, stands, colour and flicker

8
Report & exports
PDF report and aiming schedule, Relux .rdf, DXF, electrical design

What makes it different

- **It designs, not only calculates.** Press one button and it places the masts, sets their heights and the number of floodlights, aims and optimises every floodlight, then trims masts and floodlights while every check still passes – for each luminaire you tick, alone and mixed, keeping the best.
- **Scenes are checked, not assumed.** Each scene – match, training, televised, security – is calculated with only its floodlights at their levels and checked against the class set for it.
- **The checks that approvals ask for are built in:** TV camera illuminance and uniformity, glare (CIE 112), obtrusive light to neighbours and the sky (CIE 150 / EN 12464-2), stands, colour and flicker for broadcast.
- **It is verified.** A validation suite of 20 test groups, including CIE 171:2006 case 5.6, runs against every engine version (chapter 20).
- **It connects.** Masts go to the electrical design (cables, distribution), to Poles & masts (shaft and foundation), and out to Relux and CAD.

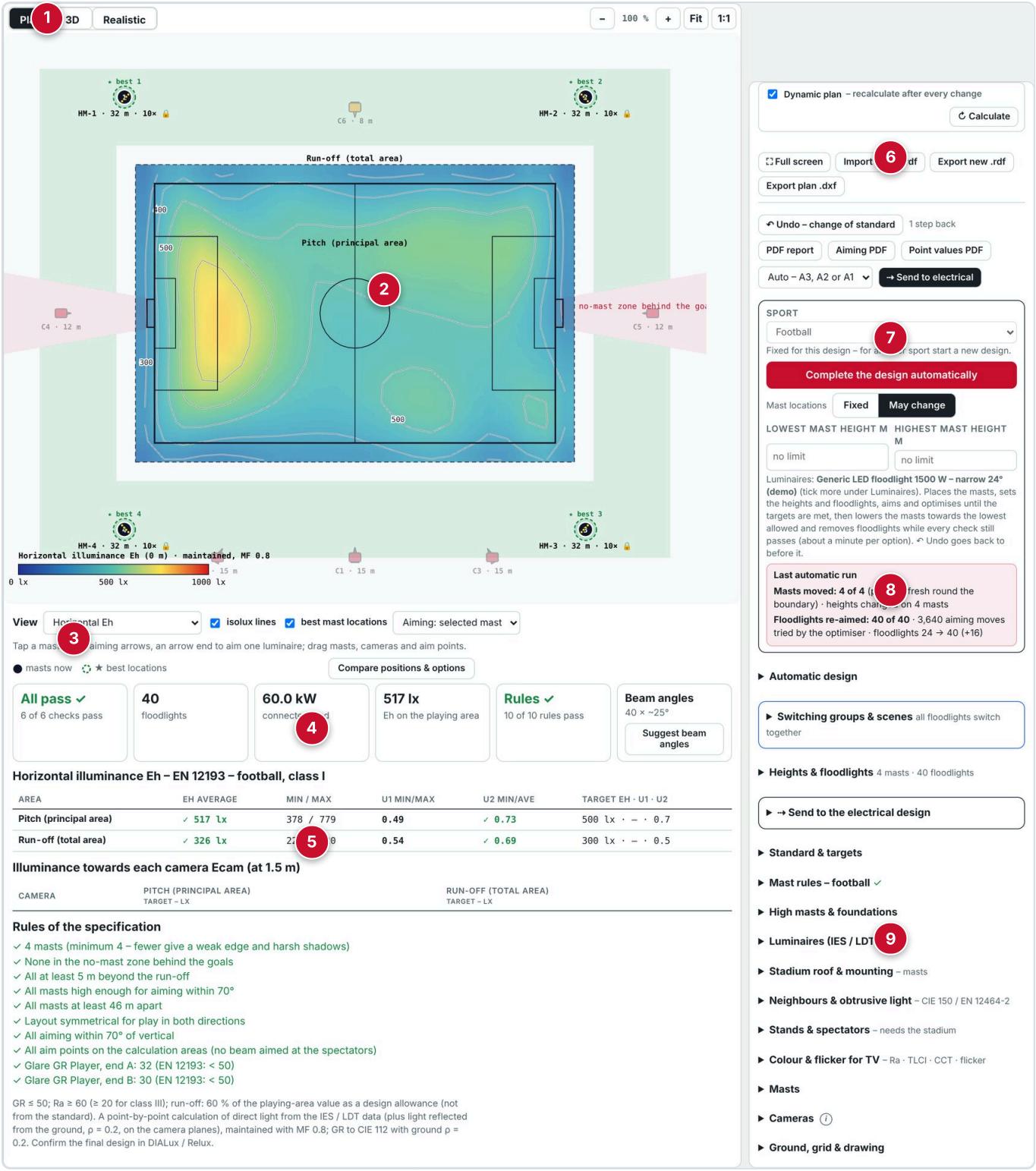
KEEP IN MIND

Light Calc is a design tool. Confirm a final design in DIALux or Relux where the client or standard asks for it – the .rdf export makes that a few minutes' work.

CHAPTER 2

The screen

Lighting → Lighting calc. The plan is on the left with the results under it; everything you set is in the side panel on the right, in panes you open as you need them.



Football, EN 12193 Class I, after the automatic design: 4 masts of 32 m, 40 floodlights, 60 kW, every check passing.

1. **Plan · 3D · Realistic** – the three views of the same design.
2. **The plan** – false colours, isolux lines, masts with height and floodlights, cameras, areas; drag to move, tap to select.

3. **View** – what the colours show (Eh, Ecam per camera, the lowest Ecam…), isolux lines, best mast locations, aiming arrows.
4. **Summary cards** – checks passing, floodlights, kW, average lux, rules, beam angles.
5. **Results** – Eh and Ecam tables against the targets, rules of the specification, glare.
6. **Toolbar** – full screen, Relux .rdf in/out, DXF, undo, PDF report, aiming and point-value PDFs, send to electrical.
7. **Sport card** – the sport and the big red **Complete the design automatically**.
8. **Last automatic run** – what the last run changed, in numbers.
9. **Panes** – Automatic design, Scenes, Heights, Standard, Mast rules, High masts, Luminaires, Stadium roof, Neighbours, Stands, Colour & flicker, Masts, Cameras, Ground.

CHAPTER 3

Calculating, undo and saving

Calculate on demand – or Dynamic plan

At the top of the side panel, **Dynamic plan** recalculates after every change. Untick it on a large design: changes then collect, a banner says *Not calculated since the last change*, and **Calculate the changes** works them all out at once. The calculation is incremental – moving one aim point recalculates one floodlight, not the design – so even with Dynamic plan on, aiming by hand stays quick.

Long runs (the automatic design, the optimiser) show their progress and a ■ **Stop** button. Stop during the automatic design puts the design back as it was before the run; Stop during the optimiser keeps the aim points moved so far.

Undo

↶ **Undo** in the toolbar names the step it will undo (*Undo – mounting*, *Undo – section type*) and counts the steps back – 30 are kept. The automatic design is one step: one Undo returns the design to before the run.

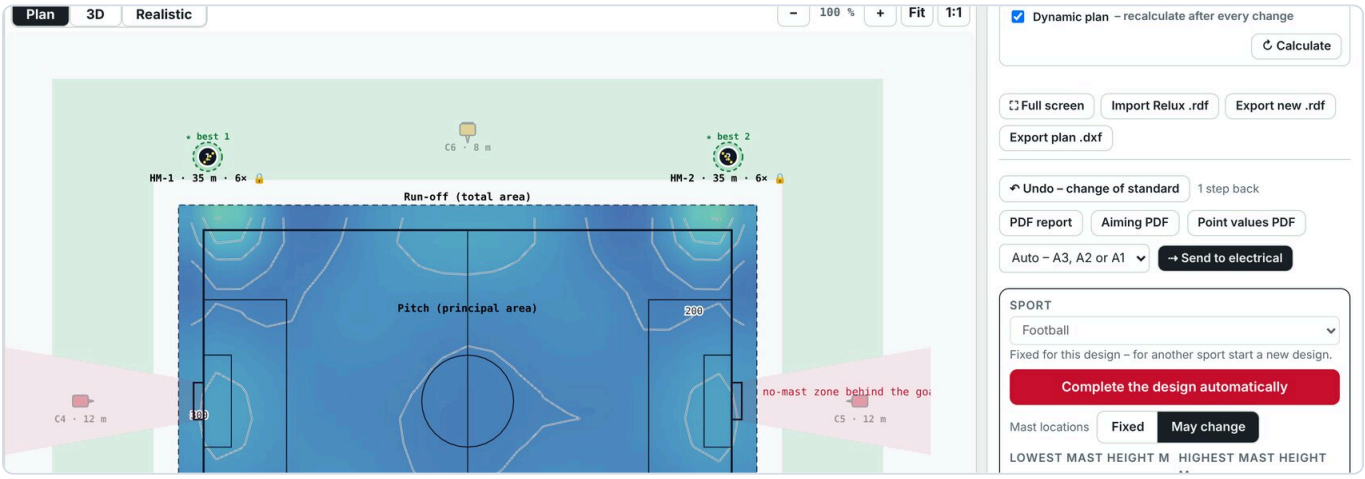
> NEW IN V155
>
> Undo also covers the stadium roof, obtrusive light, stands and colour & flicker settings.

Saving

The design is part of the project and is kept as you work – there is no Save button to forget. The PDF report carries the date and the design's revision; Relux .rdf and DXF exports are snapshots you can file with the submission.

Full screen and phones

⛶ **Full screen** gives the plan the whole window, with the side panel beside it. The tool also runs in the DIMO phone app: pinch to zoom the plan and the 3D views, tap to select, and files you export are saved through the app.



Before the first design: the football ground, the standard cameras and the green band where masts may stand, with ★ best locations.

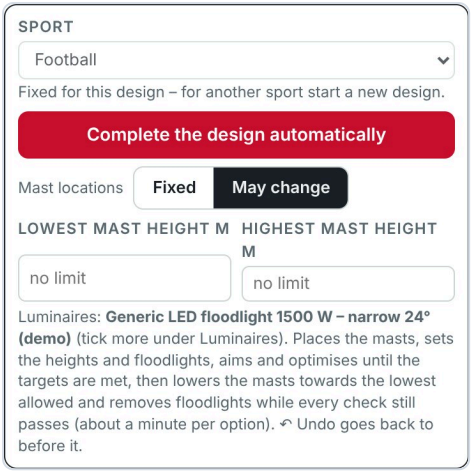
PART B

A football design, step by step

A full-size football pitch at EN 12193 Class I, from an empty ground to a passing design with scenes, DMX groups and the report. Each chapter covers one pane.

CHAPTER 4

Sport, standard and targets



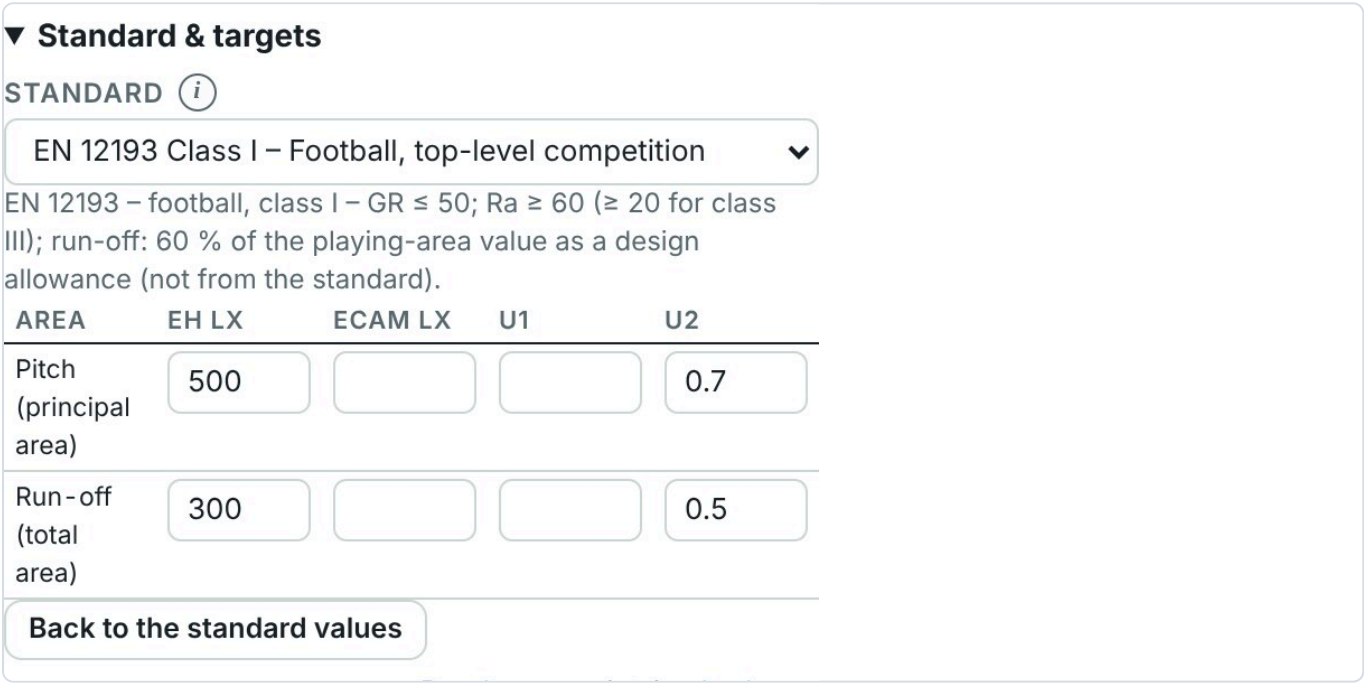
Sport card. Chosen first; fixed once the design starts.

Choose the sport first. It sets the ground, the calculation areas, the cameras, the mast rules and the standards on offer. Once the design starts the sport is fixed – another sport is a new design. For several sports on one site choose **Multi-sport facility** (Part D).

Mast locations – **May change** lets the automatic design place and move masts; **Fixed** keeps them where they are (existing masts, or masts fixed by the site). **Lowest / highest mast height** limit what the design may use – planning conditions often set a maximum.

Standards available for football

STANDARD	USE
EN 12193 Class I / II / III	top-level competition / club competition / training
Football – international broadcast (typical)	camera illuminance for TV
FIFA Class I – V	training … World-Cup-level televised



Standard & targets. The targets per area (Eh, Ecam, U1, U2) fill in from the standard and can be overtyped – **Back to the standard values** restores them. Below them the maintenance factor and ground reflection.

Maintenance factor

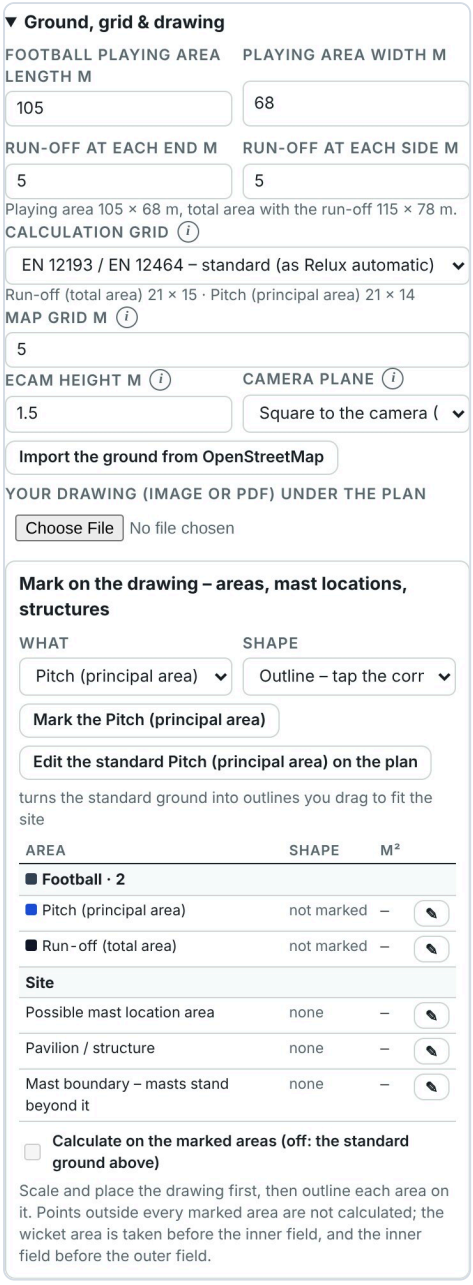
Results are maintained values – initial lux × MF. LED floodlights are usually 0.8 (L80 at end of life, clean optics); 0.7–0.75 for dusty sites or long cleaning intervals.

Light reflected from the ground

Tick to include it: grass reflects part of the light, which raises the vertical (camera) illuminance by roughly 10–15 % at ρ 0.20 but adds nothing to the horizontal. Leave it off for a conservative design; tick it to match a DIALux / Relux run that includes it.

CHAPTER 5

Ground, grid and drawing



Ground, grid & drawing.

The ground

Playing area and run-off for the sport, in metres. The calculation areas follow: for football the pitch (principal area) and the run-off (total area), each with its own targets.

Calculation grid

EN 12193 / EN 12464 – standard spaces the points as the standard asks (the same as Relux's automatic grid), so results compare one-to-one with a Relux check. A fixed spacing or your own point counts are also possible. **Map grid** sets the fineness of the colour plan only.

Cameras and Ecam

Ecam height (1.5 m usual) and **camera plane**: square to the camera (3D, as broadcasters measure) or vertical facing it.

Import the ground from OpenStreetMap

Type the ground's name or its latitude, longitude; Light Calc finds the outlines, you say which is the pitch and run-off, and it builds the areas, turned to north, with the map under the plan. Buildings round it can then be found for the obtrusive-light check (chapter 17).

Your drawing under the plan

1. Load an image or PDF of the drawing.
2. **Set the scale: tap 2 points** a known distance apart and type the distance.
3. **Move the drawing** so its pitch lies on the pitch.
4. **Mark on the drawing** – outline the pitch, run-off, possible mast locations, pavilions and structures, a mast boundary.
5. Untick **Show the drawing** once traced – it stays with the design.

> **TIP**
>
> Mark **pavilions and structures** with their height and roof shape: they then block the light in the calculation, exactly – a roof between a floodlight and the pitch casts its real shadow.

CHAPTER 6

Luminaires

Upload IES or LDT files from the maker. Each luminaire shows its lumens and watts; **Use** makes it the default for new floodlights; ✕ removes it. Four generic demo floodlights (narrow, medium, wide, and a 400 W) let you try the tool before you have files.

The Auto tick – let the design compare

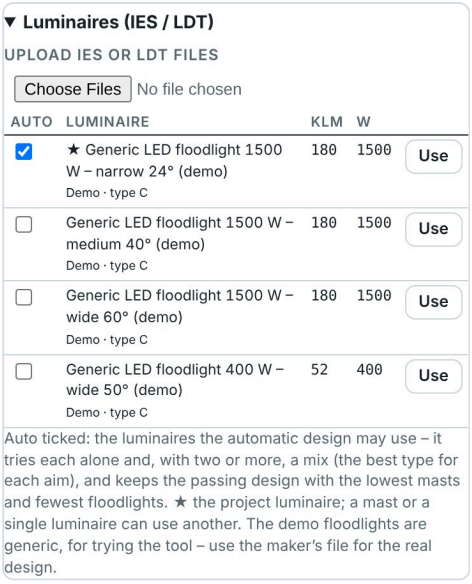
Tick every luminaire the automatic design may use. It designs with each one alone, then with all of them mixed, and keeps the passing option with the fewest floodlights and lowest masts. The others stay in memory: **Compare positions & options** shows each option's plan, results, floodlights and kW side by side.

Beam angles

The **Beam angles** card under the plan counts floodlights per beam; **Suggest beam angles** proposes narrow beams for far aim points and wider ones near the mast – mixing beams usually improves uniformity with the same count.

Colour and flicker data

CCT, Ra, R9, TLCI, flicker and frame rate per luminaire are entered under **Colour & flicker for TV** (chapter 19); CCT and Ra come in by themselves from LDT files.



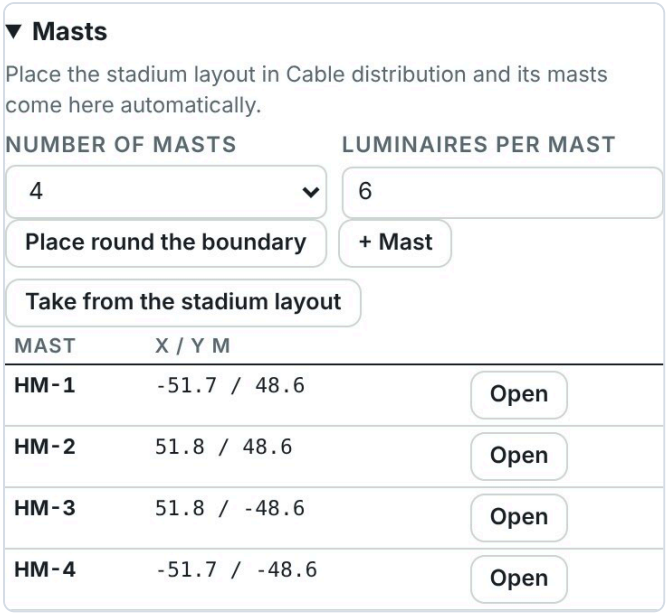
Luminaires. The ★ marks the default; ticks mark those the automatic design may use.

WATCH OUT

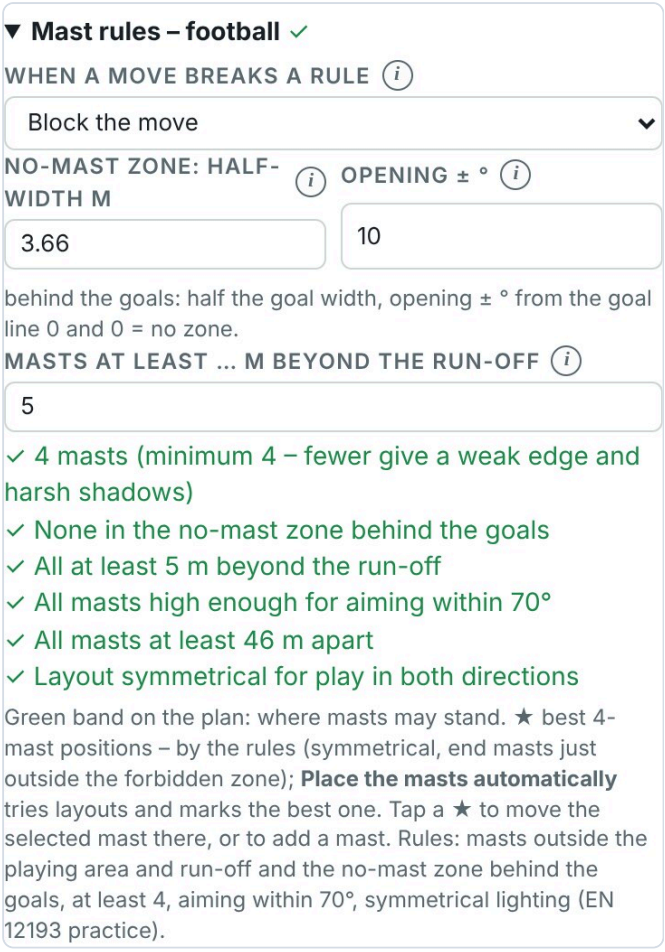
The demo floodlights are generic. A tender design needs the maker's photometry for the exact optic – beam angle alone does not describe a floodlight.

CHAPTER 7

Masts and mast rules



Masts. Number, luminaires per mast, place round the boundary, or take the masts from the stadium layout in Cable distribution.



Mast rules – football. Each rule with its status; set what happens when a move breaks one.

The rules are the sport's good practice made checkable: for football, at least 4 masts; none in the no-mast zone behind the goals (half the goal width plus an opening angle); at least 5 m beyond the run-off; high enough to aim within 70° of vertical; a minimum distance apart; and a layout symmetrical for play in both directions. **When a move breaks a rule** you choose: block the move, or allow it with a warning.

The **green band** on the plan shows where masts may stand; ★ **best locations** are the positions the tool found best for this ground and class. Tap a ★ to move the selected mast there.

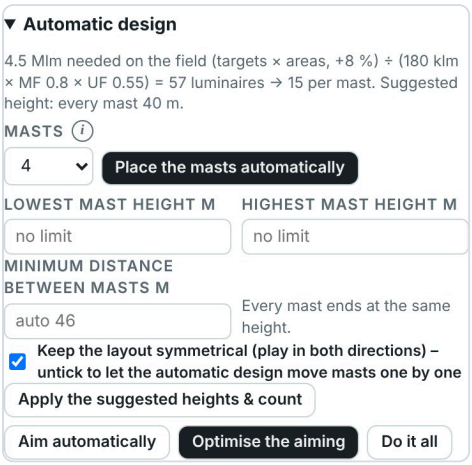
TIP - LINKED MASTS

When the stadium layout is placed in Cable distribution, tick **Masts follow the stadium layout**: the masts come here with their positions and heights and update whenever the electrical layout changes. **Send to the electrical design** sends the floodlights, kW and groups back for the cables.

CHAPTER 8

The automatic design

One button – **Complete the design automatically** – does what takes hours by hand. Here it took 28 seconds.



Automatic design. The suggestion, mast count and the steps you can run one at a time.

What it does, in order

1. **Places the masts** (if locations may change) – tries layouts within the rules and keeps the best.
2. **Sets heights and floodlight numbers** from the 25° rule, the 70° aiming limit and the lumen method.
3. **Aims and optimises** every floodlight, moving one aim point at a time while the shortfall against every target falls – up to four rounds, adding floodlights where still short.
4. **Trims a passing design** – lowers the masts towards the lowest allowed, then removes floodlights, each step kept only while every check still passes.
5. **Repeats for each ticked luminaire** and the mix, and keeps the best.
6. **With scenes**, sets each scene's level to the lowest that passes.

The message says what it did – *Design complete – every target met. 40 floodlights, masts 32 m (lowest allowed 32 m), 60.0 kW* – or, if it cannot pass, **why**: masts too low, too few, too far, or a class beyond the luminaire.

Running the steps yourself

Place the masts automatically, **Apply the suggested heights & count**, **Aim automatically**, **Optimise the aiming**, or **Do it all**. Useful when masts are fixed and you only want the aiming redone.

WATCH OUT

A must-pass broadcast scene drives the design: in the case study, requiring 1800 lx to the cameras took the stadium from 48 floodlights and 72 kW to 628 floodlights and 942 kW – and still short. Decide early whether a televised scene must pass or is reported only (chapter 12).

CHAPTER 9

Reading the results

View Horizontal Eh | isolux lines | best mast locations | Aiming: selected mast

Tap a mast for its aiming arrows, an arrow end to aim one luminaire; drag masts, cameras and aim points.

masts now ★ best locations | Compare positions & options

All pass ✓	40	60.0 kW	517 lx	Rules ✓	Beam angles
6 of 6 checks pass	floodlights	connected load	Eh on the playing area	10 of 10 rules pass	40 × ~25°
					Suggest beam angles

Horizontal illuminance Eh – EN 12193 – football, class I

AREA	EH AVERAGE	MIN / MAX	U1 MIN/MAX	U2 MIN/AVE	TARGET EH · U1 · U2
Pitch (principal area)	✓ 517 lx	378 / 779	0.49	✓ 0.73	500 lx · – · 0.7
Run-off (total area)	✓ 326 lx	226 / 420	0.54	✓ 0.69	300 lx · – · 0.5

Illuminance towards each camera Ecam (at 1.5 m)

CAMERA	PITCH (PRINCIPAL AREA) TARGET – LX	RUN-OFF (TOTAL AREA) TARGET – LX

Rules of the specification

- ✓ 4 masts (minimum 4 – fewer give a weak edge and harsh shadows)
- ✓ None in the no-mast zone behind the goals
- ✓ All at least 5 m beyond the run-off
- ✓ All masts high enough for aiming within 70°

Results after the design. Cards on top; the Eh table per area against the targets; Ecam per camera; the rules; glare per end.

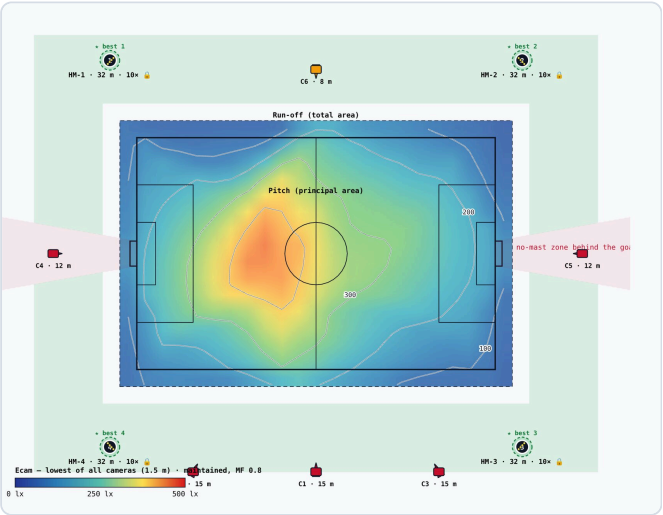
The values

- **Eh** – horizontal illuminance on the playing surface, maintained.
- **Ecam** – illuminance on a plane at 1.5 m facing each camera.
- **U1 = Emin / Emax**, **U2 = Emin / Eave** – uniformity.
- **GR** – glare rating (CIE 112) for players looking towards each mast.

✓ green passes, ✕ red fails, with the shortfall in the cell.

The views

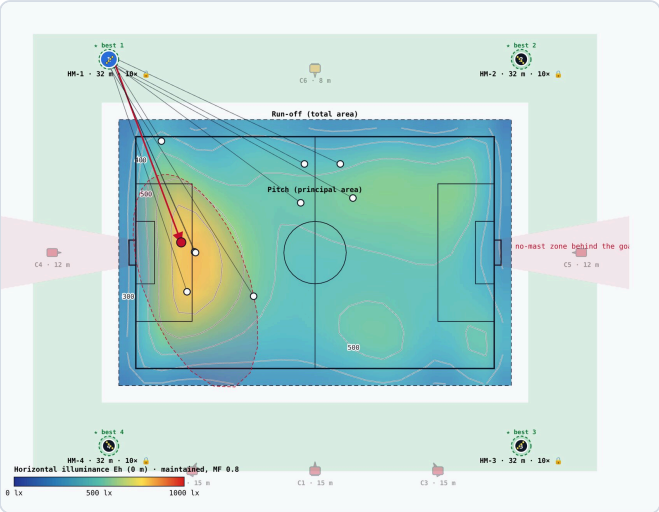
The View menu changes what the plan colours show: Eh, Ecam to any camera, the lowest Ecam of all cameras at each point (the hardest case for TV), and more. Isolux lines, the aiming arrows (none / the selected mast / all) and the best mast locations switch on and off.



Ecam – the lowest of all cameras at each point.

CHAPTER 10

Aiming by hand



One floodlight selected: its aim line in red and its 50 % beam footprint on the pitch.

1. **Tap a mast** – its aiming arrows appear.
2. **Tap an arrow end** to select one floodlight – its beam footprint shows.
3. **Drag the aim point.** The colours update as you drag (incremental calculation).
4. **Optimise** afterwards to tidy the rest around your change.

Every aim stays within 70° of vertical – a point dragged further is pulled back along its line.

When a mast moves

Each mast's **Aiming when moved** setting, shown by the symbol after its name: 🔒 aim points stay on the field; ⇄ aiming moves with the mast; ↻ re-aim automatically.

▼ **Heights & floodlights** 4 masts · 40 floodlights

HEIGHT OF ALL MASTS M	FLOODLIGHTS ON EVERY MAST
32	10

All masts +1 m | −1 m | All masts +2 floodlights | −2 | +10 %

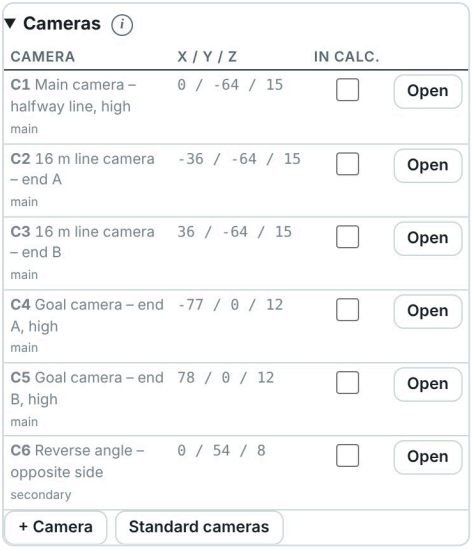
MAST	HEIGHT M	FLOODLIGHTS	FRAME	
HM-1	32	10	3.2 × 2.7 m 4 per row × 3 rows	Op
HM-2	32	10	3.2 × 2.7 m 4 per row × 3 rows	Op
HM-3	32	10	3.2 × 2.7 m 4 per row × 3 rows	Op
HM-4	32	10	3.2 × 2.7 m 4 per row × 3 rows	Op

Changing a count keeps the aiming – new floodlights copy the existing aim points. Then optimise the aiming. ↶ Undo goes back one step.

Heights & floodlights. All masts at once (+1 m, +2 floodlights, +10 %) or each mast; its frame size and rows. New floodlights copy the existing aim points – optimise after.

CHAPTER 11

TV cameras

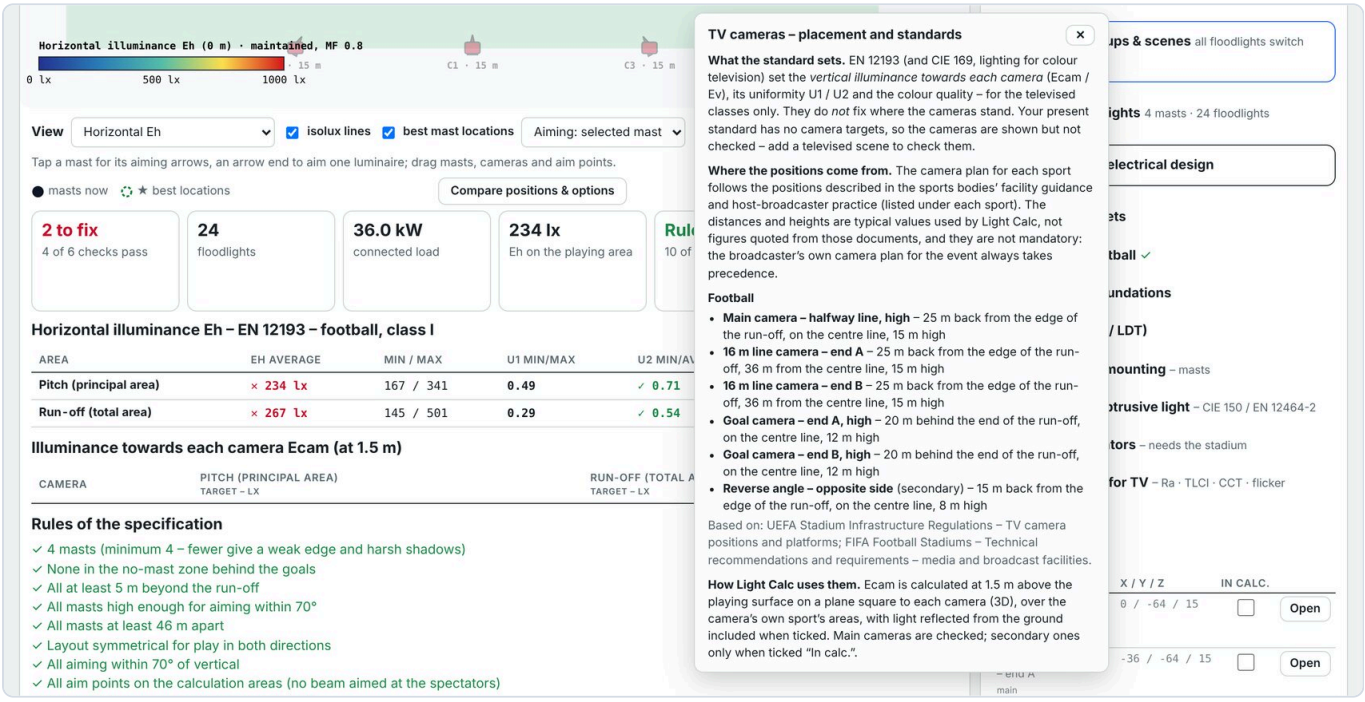


Cameras. Standard positions for the sport, each ticked in or out of the calculation.

Standard cameras places the usual camera plan for the sport: main camera on the halfway line, 16 m line cameras, goal cameras, reverse angle. Tick a camera **In calc.** and the Ecam values towards it appear in the results.

Drag a camera and it shows ⚠ **moved from the standard position**; ⟲ **Standard position** beside its Open button puts it back – one button puts back all moved cameras.

The **i** next to Cameras explains the placement, the standards behind it, where the positions come from, and how Light Calc uses them.



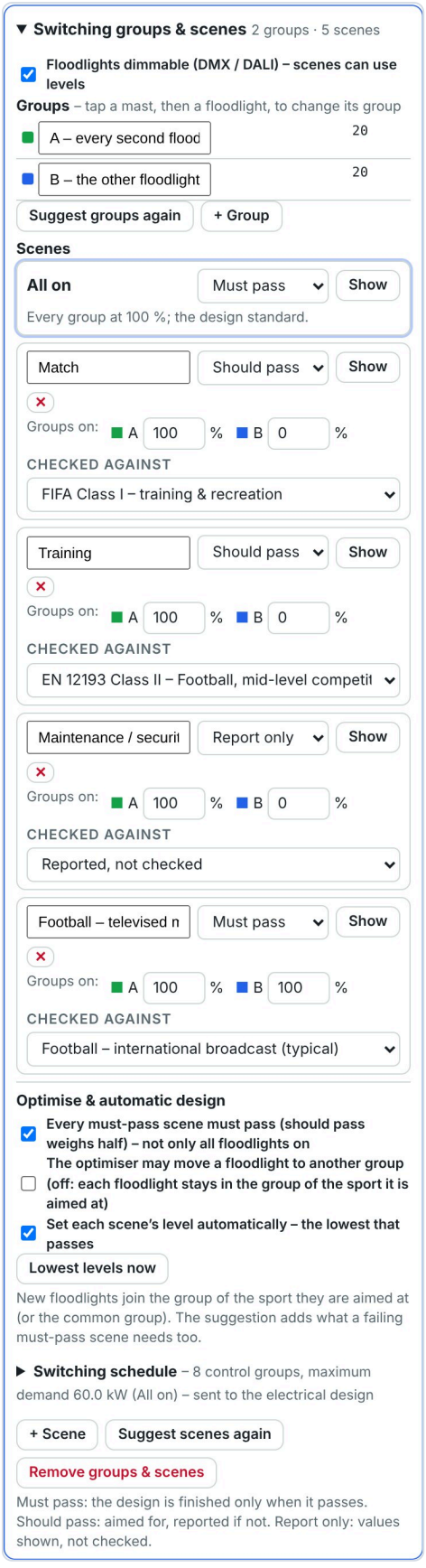
The camera i. What the standards ask, where the positions come from, how Ecam is calculated.

CAMERAS COUNT IN TELEVISED SCENES

A standard without camera targets (EN 12193 Class I–III) does not check the cameras. Light Calc points this out and offers two ways: a televised scene that must meet a broadcast class, or a report-only scene giving the broadcaster the camera values from the present lighting.

CHAPTER 12

Switching groups, scenes and DMX



Switching groups & scenes.

Groups

Floodlights are put into control groups – suggested as *A – every second floodlight* and *B – the other*, or by mast or sport. Tap a mast, then a floodlight, to change its group. Tick **dimmable (DMX / DALI)** and scenes can use levels; untick for contactors.

Scenes

Each scene sets every group's level and the class it is checked against, with a requirement:

- **Must pass** – the design is finished only when it passes.
- **Should pass** – aimed for, reported if not.
- **Report only** – values shown, not checked (security, present-lighting TV).

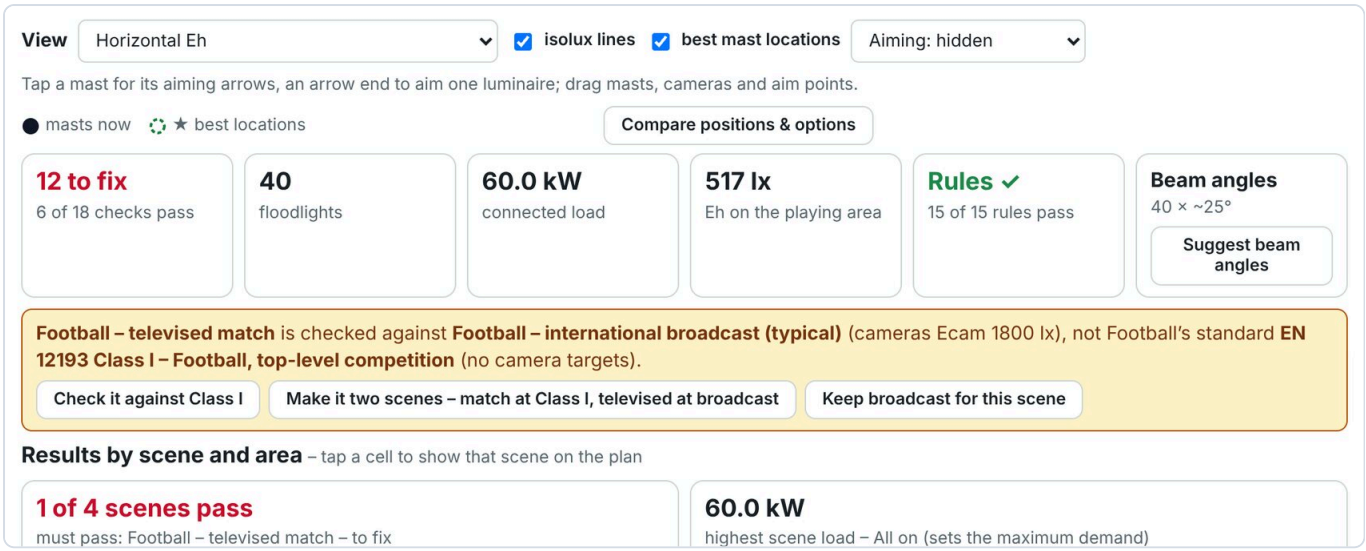
Suggest scenes creates the usual set for the sport: match, training, televised, maintenance / security.

Optimise and automatic design with scenes

Options: every must-pass scene must pass; the optimiser may move a floodlight to another group; set each scene's level automatically to the lowest that passes.

Switching schedule

Control groups with their floodlights and kW, each scene's groups and levels, and the **maximum demand** – the highest scene load, which sizes the supply.



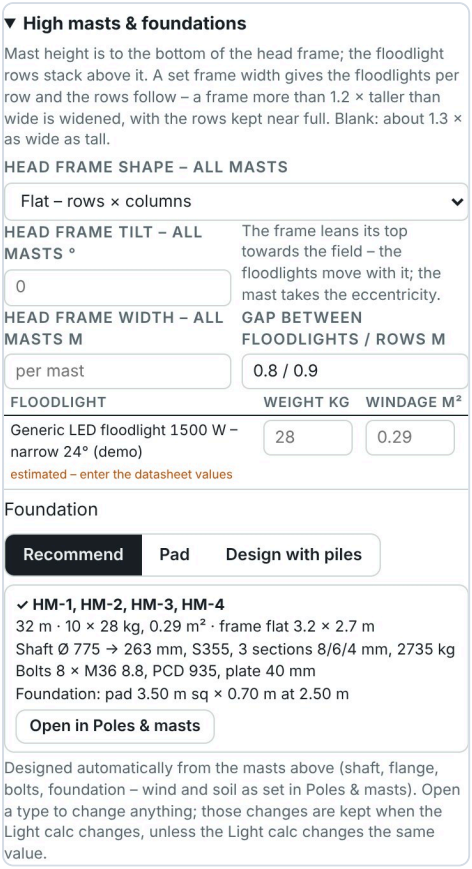
Results by scene. Scenes added after the design: *Match* and *Training* now switch only half the floodlights and fail. The yellow banner spots that the televised scene is checked against a broadcast class, not the club standard, and offers three ways out. Pressing **Complete the design automatically** again designs every scene together.

TIP – FOLLOW THE STANDARD

Change the sport's standard and the scenes that used it follow. A must-pass scene set to a higher class than the standard is flagged with the way back.

CHAPTER 13

High masts and foundations



High masts & foundations.

Each mast type gets a structural design from its height and head load:

- **Head frame** – shape (flat, stepped, curved, V, trapezoid, ring), tilt, width and gaps; floodlights per row and rows follow.
- **Floodlight weight and windage** – from the maker, or estimated from the wattage (marked as estimated).
- **Shaft** – diameter at the base and top, steel grade, sections and thicknesses, mass.
- **Base** – bolts, PCD, plate.
- **Foundation** – recommended, pad, or piles, with sizes.

The wind and soil come from Poles & masts; **Open in Poles & masts** shows the full calculation. Changes made there are kept for that type.

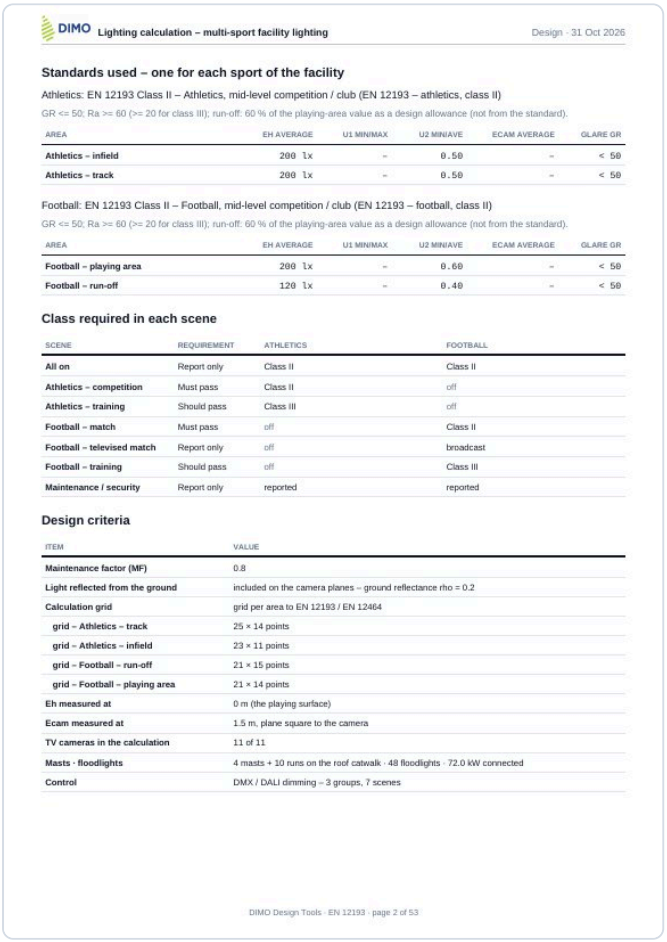
WATCH OUT

Enter the maker's weight and windage. Estimates are marked in red until you do, and the foundation follows them.

CHAPTER 14

Reports and exports

BUTTON	WHAT YOU GET
PDF report	Summary cards and plan; the standards used (every sport's in a multi-sport design) and design criteria; Eh, Ecam, U1/U2 and glare tables; every scene with its plan and results; obtrusive light, stands, colour & flicker when used; luminaires; masts and roof sections, each head frame drawn with every floodlight numbered and coloured by group and its level in each scene; DMX control groups.
Aiming PDF	One section per mast: its aiming plan and every floodlight's aim point, tilt, direction and distance – for the installer.
Point values PDF	Every calculation point's values – for an approving engineer.
Page size	Auto, A3, A2 or A1 for the plans.
Import Relux .rdf	Opens a Relux project – floodlights as built, with their photometry and aiming.
Export new .rdf	A new Relux project with every floodlight, aiming, photometry, the ground and the main camera – no Relux file of your own needed.
Export plan .dxf	Ground lines, areas, masts, aiming lines, cameras, stands, roof and catwalk on their own layers, at the export position.
Send to electrical	Masts, floodlights, kW and control groups to the electrical design.



DIMO Lighting calculation – multi-sport facility lighting — Design · 31 Oct 2026

Standards used – one for each sport of the facility

Athletics: EN 12193 Class II – Athletics, mid-level competition / club (EN 12193 – athletics, class II)

GR <= 50; Ra >= 60 (>= 20 for class III); run-off: 60 % of the playing-area value as a design allowance (not from the standard).

AREA	EH AVERAGE	U1 MIN/MAX	U2 MIN/AVE	ECAM AVERAGE	GLARE GR
Athletics – infield	200 lx	–	0.50	–	< 50
Athletics – track	200 lx	–	0.50	–	< 50

Football: EN 12193 Class II – Football, mid-level competition / club (EN 12193 – football, class II)

GR <= 50; Ra >= 60 (>= 20 for class III); run-off: 60 % of the playing-area value as a design allowance (not from the standard).

AREA	EH AVERAGE	U1 MIN/MAX	U2 MIN/AVE	ECAM AVERAGE	GLARE GR
Football – playing area	200 lx	–	0.60	–	< 50
Football – run-off	120 lx	–	0.40	–	< 50

Class required in each scene

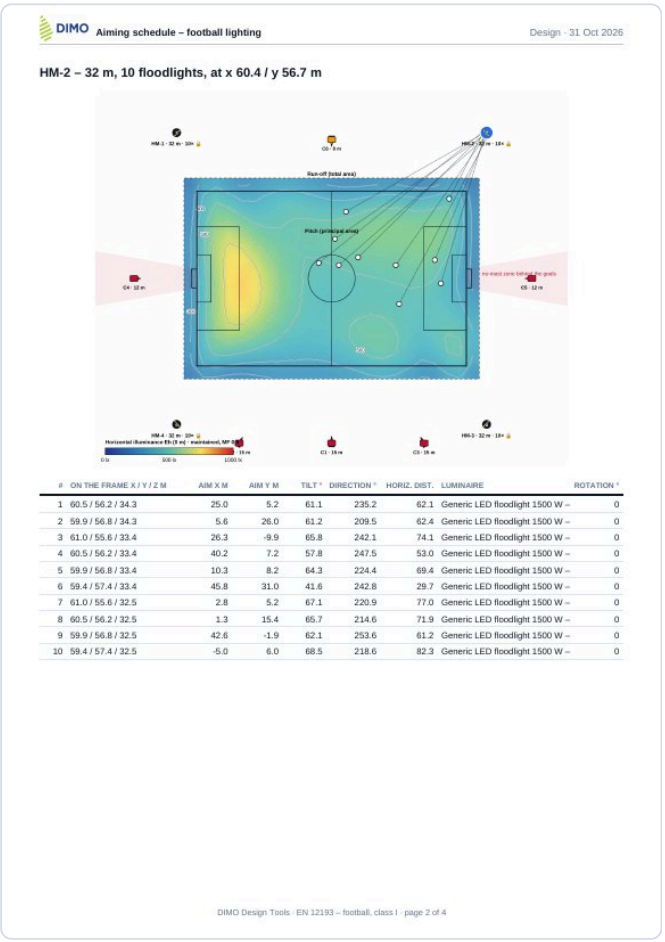
SCENE	REQUIREMENT	ATHLETICS	FOOTBALL
All on	Report only	Class II	Class II
Athletics – competition	Must pass	Class II	off
Athletics – training	Should pass	Class III	off
Football – match	Must pass	off	Class II
Football – televised match	Report only	off	broadcast
Football – training	Should pass	off	Class III
Maintenance / security	Report only	reported	reported

Design criteria

ITEM	VALUE
Maintenance factor (MF)	0.8
Light reflected from the ground	included on the camera planes – ground reflectance rho = 0.2
Calculation grid	grid per area to EN 12193 / EN 12464
grid – Athletics – track	25 × 14 points
grid – Athletics – infield	23 × 11 points
grid – Football – run-off	21 × 15 points
grid – Football – playing area	21 × 14 points
Eh measured at	0 m (the playing surface)
Ecam measured at	1.5 m, plane square to the camera
TV cameras in the calculation	11 of 11
Masts · floodlights	4 masts + 10 runs on the roof catwalk · 48 floodlights · 72.0 kW connected
Control	DMX / DALI dimming – 3 groups, 7 scenes

DIMO Design Tools · EN 12193 · page 2 of 53

Report: standards used, one per sport.



DIMO Aiming schedule – football lighting — Design · 31 Oct 2026

HM-2 – 32 m, 10 floodlights, at x 60.4 / y 56.7 m

#	ON THE FRAME X / Y / Z M	AIM X M	AIM Y M	TILT °	DIRECTION °	HORIZ. DIST.	LUMINAIRE	ROTATION °
1	60.5 / 56.2 / 34.3	25.0	5.2	61.1	235.2	62.1	Generic LED floodlight 1500 W –	0
2	59.9 / 56.8 / 34.3	5.6	26.0	61.2	209.5	62.4	Generic LED floodlight 1500 W –	0
3	61.0 / 55.6 / 33.4	26.3	-9.9	65.8	242.1	74.1	Generic LED floodlight 1500 W –	0
4	60.5 / 56.2 / 33.4	40.2	7.2	57.8	247.5	53.0	Generic LED floodlight 1500 W –	0
5	59.9 / 56.8 / 33.4	10.3	8.2	64.3	224.4	69.4	Generic LED floodlight 1500 W –	0
6	59.4 / 57.4 / 33.4	45.8	31.0	41.6	242.8	29.7	Generic LED floodlight 1500 W –	0
7	61.0 / 55.6 / 32.5	2.8	5.2	67.1	220.9	77.0	Generic LED floodlight 1500 W –	0
8	60.5 / 56.2 / 32.5	1.3	15.4	65.7	214.6	71.9	Generic LED floodlight 1500 W –	0
9	59.9 / 56.8 / 32.5	42.6	-1.9	62.1	253.6	61.2	Generic LED floodlight 1500 W –	0
10	59.4 / 57.4 / 32.5	-5.0	6.0	68.5	218.6	82.3	Generic LED floodlight 1500 W –	0

DIMO Design Tools · EN 12193 – football, class I · page 2 of 4

Aiming PDF: one mast's aiming plan and schedule.

PART C

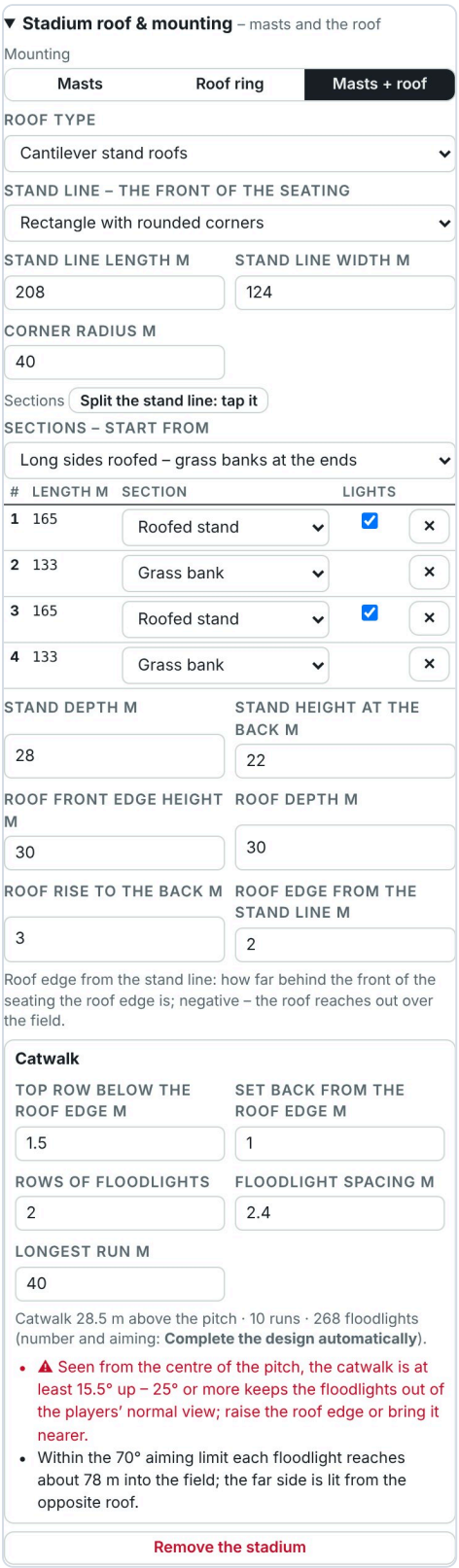
The powerful features

Roof-mounted floodlights, the photo-real view, the checks planning and broadcasters ask for, and how far you can trust the numbers.

CHAPTER 15

Stadium roof and roof mounting

Draw the stadium round the field – stands, grass banks, the roof – and mount floodlights on the roof catwalk, on masts, or both.



Stadium roof & mounting.

Mounting

Masts – floodlights on masts; the stadium is drawn and its roofs shade the masts' light. **Roof ring** – floodlights on the roof catwalk. **Masts + roof** – both. Your masts are kept when you switch.

Roof type

Cantilever, column-supported canopy, continuous ring (bowl), cable-net, arch-supported, retractable, dome, segmented curved cantilever. Ring, cable-net, retractable and dome roofs are carried all round, so every section is roofed.

Stand line and sections

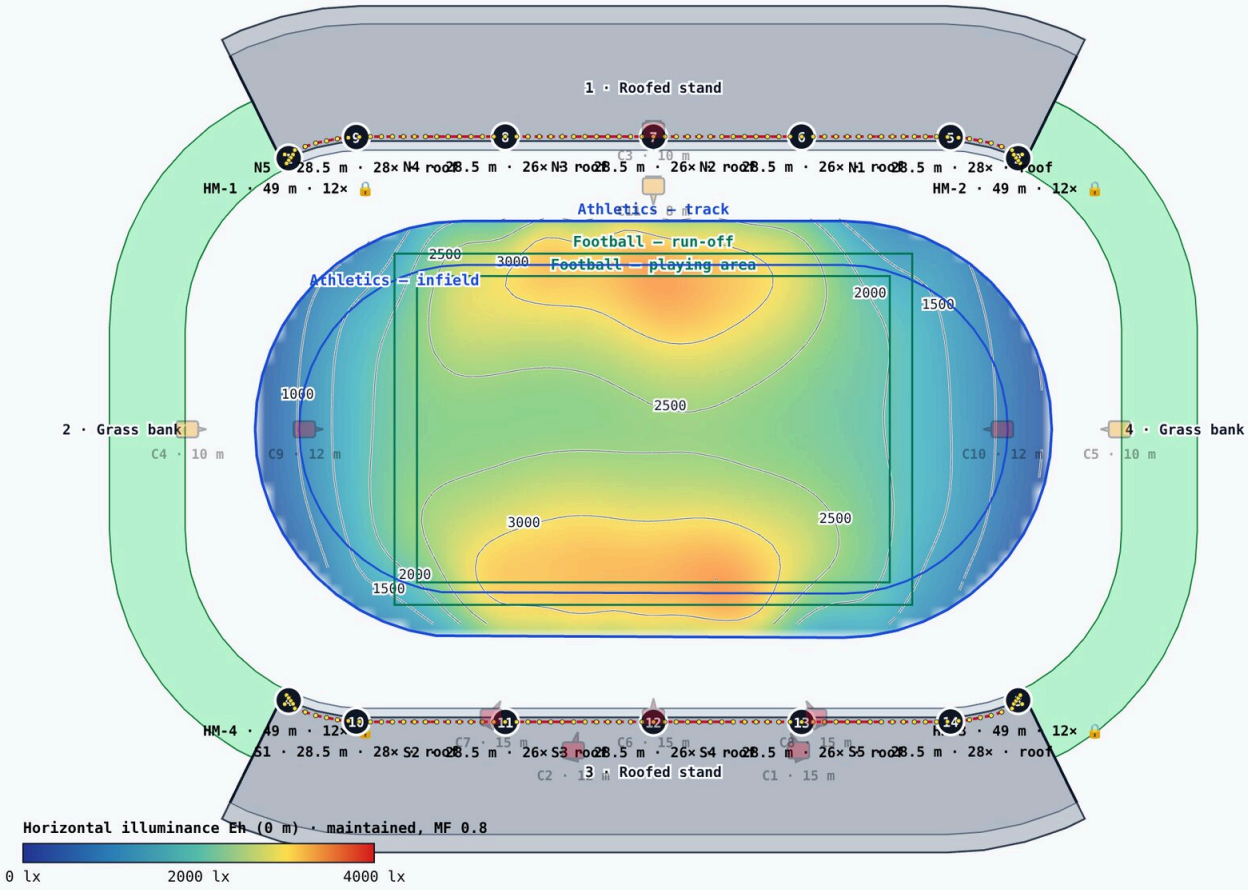
The **stand line** is the front of the seating: a rounded rectangle, an oval, or **traced on the plan** (tap round it on your drawing). It is cut into sections – roofed stand, open stand, grass bank or none – from a preset or by tapping the line. Each roofed section can carry floodlights or not.

Sizes and the catwalk

Stand depth and height; roof front-edge height, depth, rise and edge position. Catwalk: top row below the roof edge, set back, rows (1–3), spacing, longest run.

Runs work like masts

The catwalk is cut into runs (N1, N2 … S1 …). Each run is designed, aimed, optimised, grouped and reported exactly like a mast; its height stays the catwalk height.



On the plan: stands (grey), grass banks (green), roofs, the catwalk runs (red dashes) and four corner masts – Masts + roof.

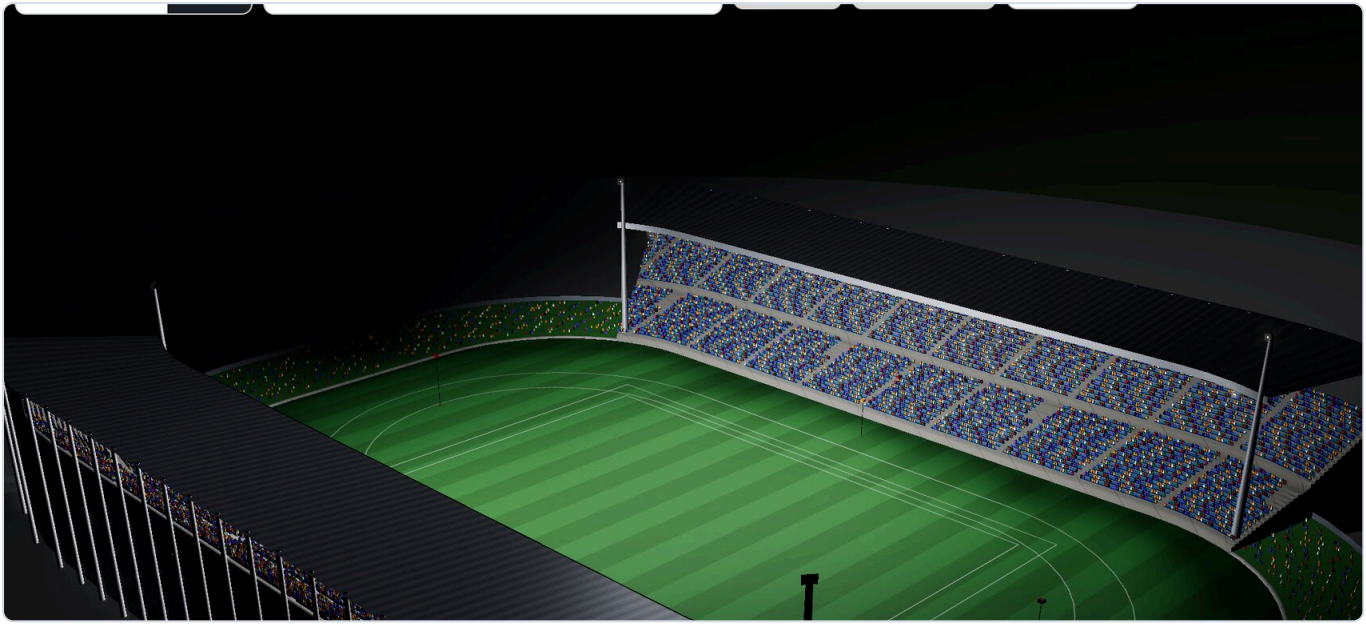
CHECKS FOR ROOF MOUNTING

The 25° line from the centre of the pitch to the catwalk (lower floodlights glare into the players' view), and how far each floodlight reaches within the 70° aiming limit – the opposite roof lights the far side.

CHAPTER 16

The realistic view

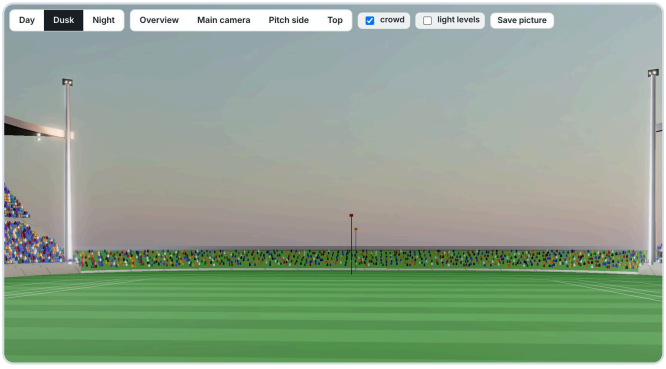
Plan · 3D · **Realistic**: the stadium, masts and floodlights as they will look – sky and sun, materials, seats and crowd, the roof by type, every floodlight aimed as designed.



Night. The pitch brightness comes from the calculated illuminance – a club-level pitch looks dimmer than a broadcast one. The caption gives floodlights, average lux and seats.



Day – for the architectural picture.



Dusk, pitch side.

Controls: Day · Dusk · Night; Overview, Main camera, Pitch side, Top; crowd on or off; light-levels overlay; **Save picture** (PNG for proposals). Drag to turn, scroll or pinch to zoom, right-drag to move.

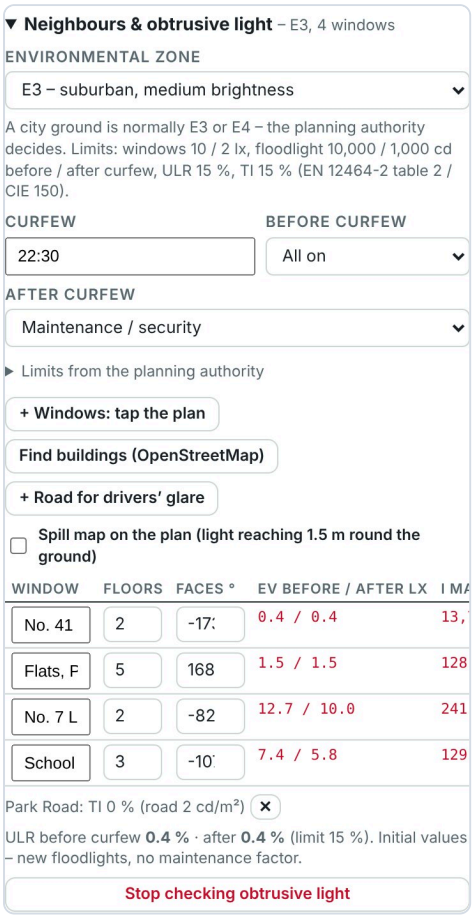
KEEP IN MIND

The view loads its 3D library the first time it opens, so it needs an internet connection then; Plan and 3D work offline. Only the pitch brightness is calculated – light on the stands and roof is for looks. Turn the crowd off on older phones.

CHAPTER 17

Neighbours and obtrusive light

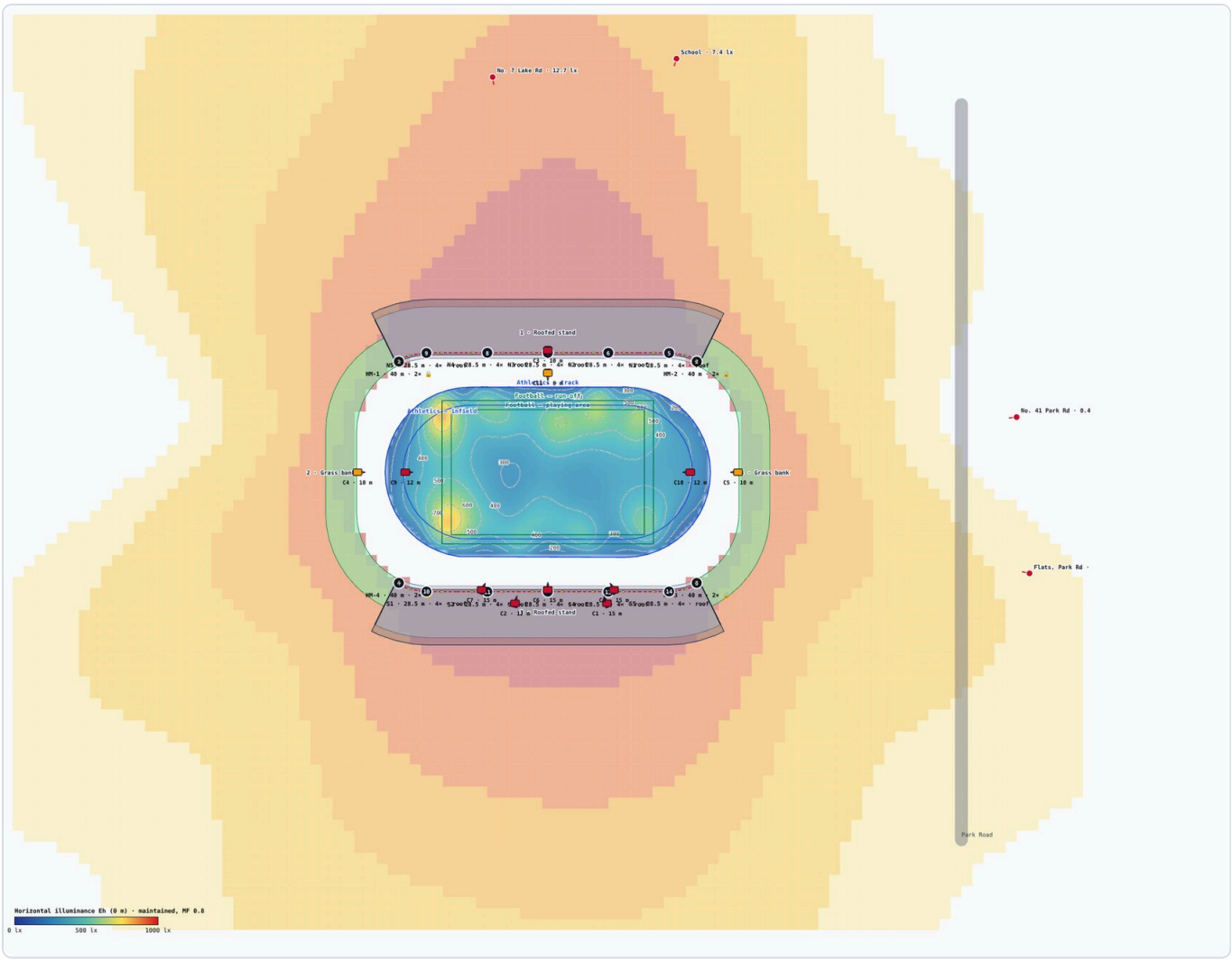
The check planning approval asks for near housing: light on neighbours' windows, the brightest floodlight they see, light going up into the sky, and glare for drivers – before and after the curfew, against the limits of the area's environmental zone (CIE 150 / EN 12464-2).



Neighbours & obtrusive light.

1. **Check obtrusive light**, then the **environmental zone** – E1 natural … E4 city centre. The planning authority decides; a city ground is usually E3 or E4. Its own figures can be typed instead.
2. **Curfew** and the scenes **before** and **after** it (usually match before; security or all off after).
3. **Windows** – tap the plan, or **Find buildings (OpenStreetMap)**: the 30 nearest buildings each get a window on the façade facing the ground, floors from the map. Set floors and direction in the table.
4. **Road** – tap along a road for drivers' glare (TI), both directions.
5. Optional **spill map** – light reaching 1.5 m round the ground.

Each window is calculated floor by floor like a camera plane – every floodlight, photometry and aim, with roofs blocking. Values are **initial** (new floodlights) – the worst case.



Spill map and windows on the plan – each window coloured pass / close / over, with its lux.

FIXING A FAILING WINDOW

Aim the floodlights that face it a few degrees lower; fit the maker's rear visor (load the visor's photometry); use an asymmetric optic; or lower the after-curfew scene. Each change shows on the window straight away.

CHAPTER 18

Stands and spectators

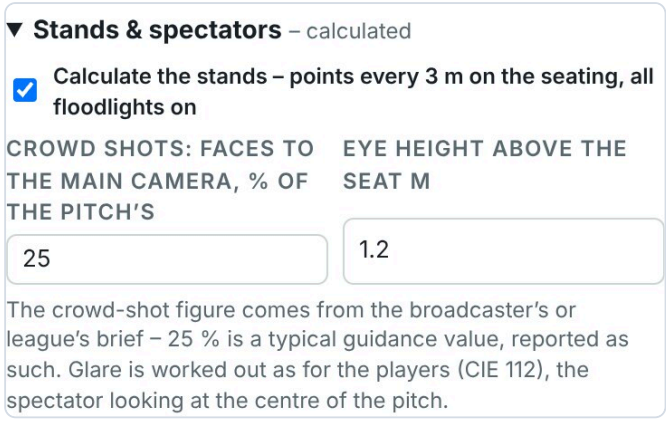
With the stadium drawn, tick **Calculate the stands**: points every 3 m on the seating, all floodlights on. For each section:

- **Eh on the seats**, average, minimum and U1.
- **Faces to the main camera** – vertical illuminance at eye height towards the camera, as a share of the pitch's. Stands behind the camera are marked *back to it*.
- **GR** – glare for a spectator looking at the centre of the pitch (CIE 112).

The crowd-shot share (default 25 %) comes from the broadcaster's or league's brief – it is guidance, not a standard requirement, and the report says so.

Roofs cutting the masts' beams

With masts behind roofed stands, each floodlight's 50 % beam is traced against the roof planes. Those losing 5 % or more are listed, with the mast height that clears them – or *re-aim* when none within 25 m does.



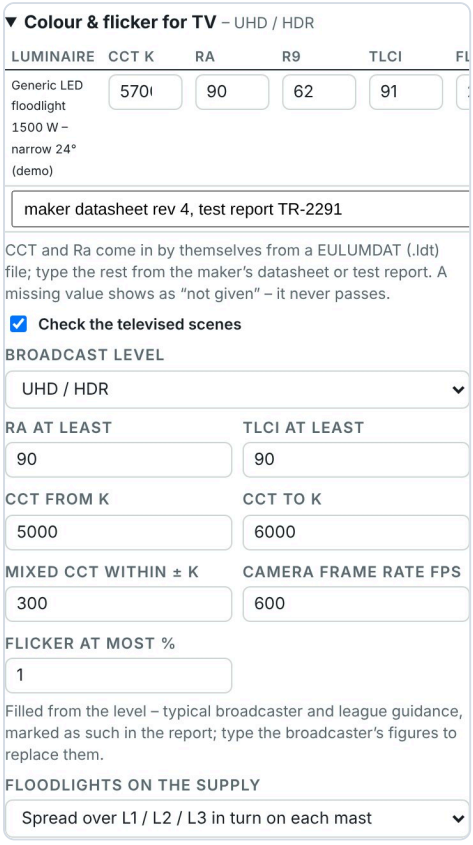
Stands & spectators.



Report page – stands, with the case study's values.

CHAPTER 19

Colour and flicker for TV



Colour & flicker for TV.

Per luminaire

CCT, Ra, R9, TLCI, flicker % and the camera frame rate it is flicker-free to, with the source (datasheet, test report). CCT and Ra come in from LDT files. A missing value shows as *not given* – it never passes.

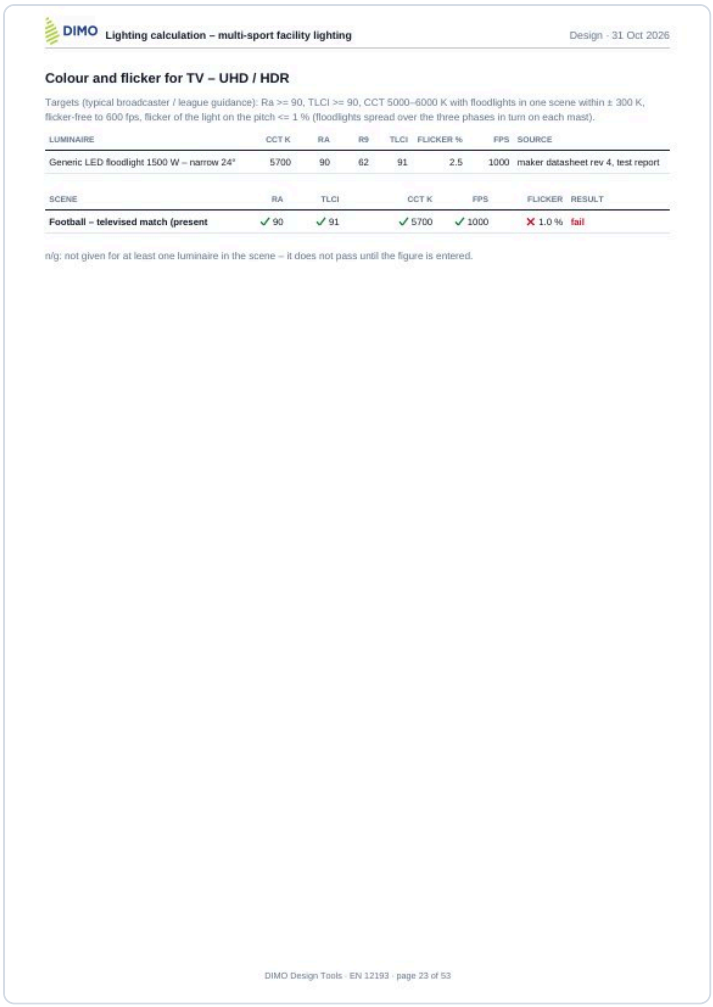
Per televised scene

Pick the broadcast level – **HD**, **UHD / HDR**, **super slow-motion** – and its targets fill in (editable; typical broadcaster and league guidance, marked as such in the report):

LEVEL	RA	TLCI	CCT K	± K	FPS	FLICKER
HD	80	80	4000–6500	500	300	3 %
UHD / HDR	90	90	5000–6000	300	600	1 %
Super slow-motion	90	90	5000–6000	300	1000	1 %

Combined flicker on the pitch

Floodlights on different phases ripple at different times and largely cancel. Light Calc adds each floodlight's ripple as a vector at each point of the pitch – floodlights spread over L1 / L2 / L3 in turn on each mast, or all on one phase as the worst case – and reports the highest.



DIMO Lighting calculation – multi-sport facility lighting

Design · 31 Oct 2026

Colour and flicker for TV – UHD / HDR

Targets (typical broadcaster / league guidance): Ra >= 90, TLCI >= 90, CCT 5000–6000 K with floodlights in one scene within ± 300 K, flicker-free to 600 fps, flicker of the light on the pitch <= 1 % (floodlights spread over the three phases in turn on each mast).

LUMINAIRE	CCT K	RA	R9	TLCI	FLICKER %	FPS	SOURCE
Generic LED floodlight 1500 W – narrow 24°	5700	90	62	91	2.5	1000	maker datasheet rev 4, test report

SCENE	RA	TLCI	CCT K	FPS	FLICKER	RESULT
Football – televised match (present	✓ 90	✓ 91	✓ 5700	✓ 1000	✗ 1.0 %	fail

n/g: not given for at least one luminaire in the scene – it does not pass until the figure is entered.

DIMO Design Tools · EN 12193 · page 23 of 53

Report page – the case study's televised scene: Ra and TLCI pass; the combined flicker of 1.01 % is just over the 1 % UHD target.

CHAPTER 20

Accuracy and validation

Every engine version is run against a suite of analytical test cases in the style of CIE 171:2006 – cases with exact answers worked out independently of the engine's code.

#	TEST	LIMIT
1–3	Point source: inverse-square and cosine laws; vertical planes; planes square to a TV camera	0.1 %
4–7	Aimed floodlights at 50–70° tilt; asymmetric distributions and spin; table interpolation; flux and IES = LDT	0.01–1 %
8	Four masts summed with the maintenance factor (full calculation)	1 %
9a/9b	Ground reflection onto camera planes – against fine numerical integration	5 / 10 %
10	Glare rating GR (CIE 112)	0.1 %
11/11b	Light blocked by stands – exact line-of-sight	0.01 %
12	Energy balance – flux reaching the ground = flux sent down	1 %
13	CIE 171:2006 case 5.6 – published reference values	5 %
14	Roof catwalk positions; roof shading a mast	0.5 %
15–17	Neighbour's window Ev and intensity; upward light ratio; drivers' TI	0.1–0.5 %
18	Combined flicker on three phases	0.01 %

All 20 groups pass on v155. The calculation itself: point-by-point direct light from the IES / LDT tables (bilinear between table values), ground reflection on the camera planes with near-field rings and far-field patches, glare to CIE 112 within 1.5°–60°, and exact obstruction by structures and roofs.

INDEPENDENT CHECK

For a submission, export the design as a new Relux .rdf and run it in Relux – the grid follows the same standard spacing, so the tables compare directly.

CHAPTER 21

Twelve power tips

1. **Tick two or three luminaires for the automatic design.** It compares each and the mix, and Compare positions & options shows them side by side.
2. **Untick Dynamic plan on big designs** and press Calculate the changes when ready.
3. **Set the scenes before the automatic design** – it then designs every scene and sets each one's lowest passing level.
4. **Decide early: must-pass or report-only TV.** It can multiply the floodlights several times.
5. **Use ★ best locations** for the first layout, then let the rules guard your moves.
6. **Fixed masts?** Set Mast locations to Fixed and use Aim automatically + Optimise.
7. **Mix beams** with Suggest beam angles before adding floodlights.
8. **Mark structures** on the drawing – their shadows are exact.
9. **Link the masts to the stadium layout** and Send to electrical, so cables and lighting never drift apart.
10. **Run obtrusive light early** near housing – fixing aim and visors is cheap before the masts are ordered.
11. **Save a realistic picture** at night from the main camera for the proposal.
12. **Export a new .rdf** for the independent Relux check before submission.

PART D – CASE STUDY

A multi-sport stadium

Athletics and football on one ground, roofed long sides and grass banks at the ends, floodlights on corner masts and the roof catwalk, houses and a school nearby, and a broadcaster asking for camera values. Worked through in Light Calc from start to finish.

CHAPTER 22

The brief

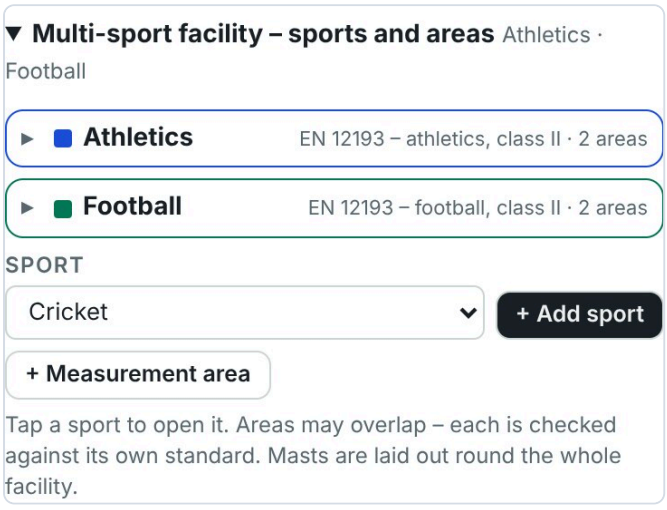
“Colombo City Sports Complex” – a fictional ground used for this example. The site, houses and luminaire data are invented; every result is Light Calc’s own.

ITEM	BRIEF
Sports	400 m athletics track, 8 lanes, with a football pitch inside
Standards	Athletics: EN 12193 Class II (club competition). Football: EN 12193 Class II
Scenes	Athletics competition (must pass), athletics training, football match (must pass), football training, televised football – camera values for the broadcaster (report only), maintenance / security
Stadium	Roofed stands on both long sides (cantilever roof, front edge 30 m); grass banks at both ends
Mounting	Masts + roof: four corner masts behind the grass banks, plus the roof catwalk over the long sides
Control	DMX dimming
Neighbours	Houses and flats on Park Road (east, about 255 m), houses and a school on Lake Road (north, about 215 m); zone E3; curfew 22:30
TV	UHD coverage of football; floodlight data: CCT 5700 K, Ra 90, R9 62, TLCI 91, flicker 2.5 %, flicker-free to 1000 fps (example values)

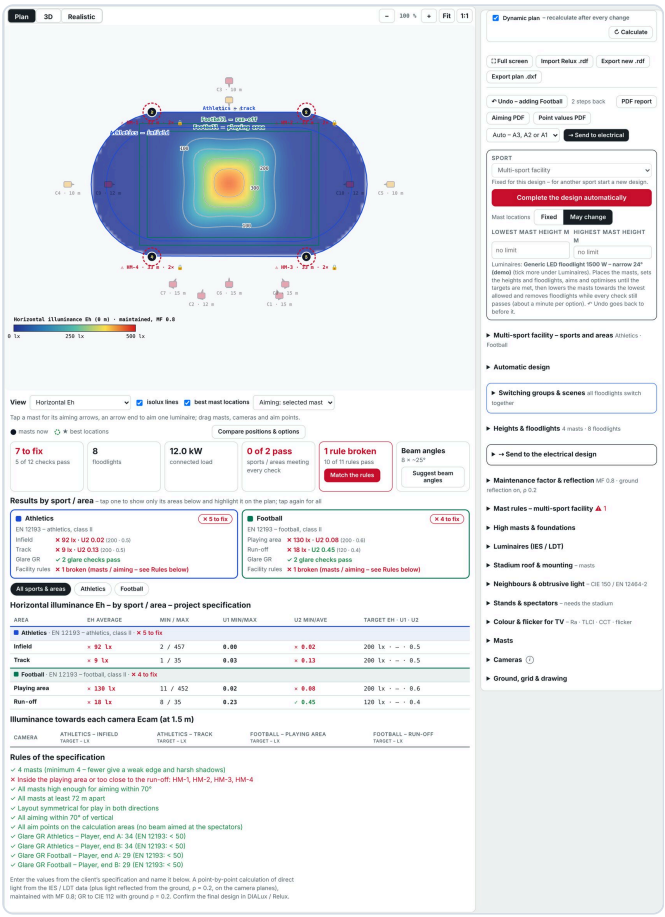
CHAPTER 23

Setting it up

1. **Sport: Multi-sport facility.** Add **Athletics**, then **Football**; centre the pitch on the track. Each sport gets its own areas – athletics infield and track, football playing area and run-off – and its own standard.
2. **Standards:** athletics EN 12193 Class II, football EN 12193 Class II, each set in its sport's line.
3. **Stadium:** stand line 208 × 124 m, corners 40 m; **Sections – Long sides roofed – grass banks at the ends**; cantilever roof, front edge 30 m, depth 30 m, stands 28 m deep.
4. **Masts:** four corner masts placed round the boundary; then **Mounting: Masts + roof**. The catwalk is cut into ten runs, N1–N5 and S1–S5, at 28.5 m.
5. **Scenes:** Suggest scenes, then **Report the cameras only – broadcast with the present lighting** for football; DMX ticked.



Sports and areas – each sport with its standard.



The ground – track, infield and pitch.

CHAPTER 24

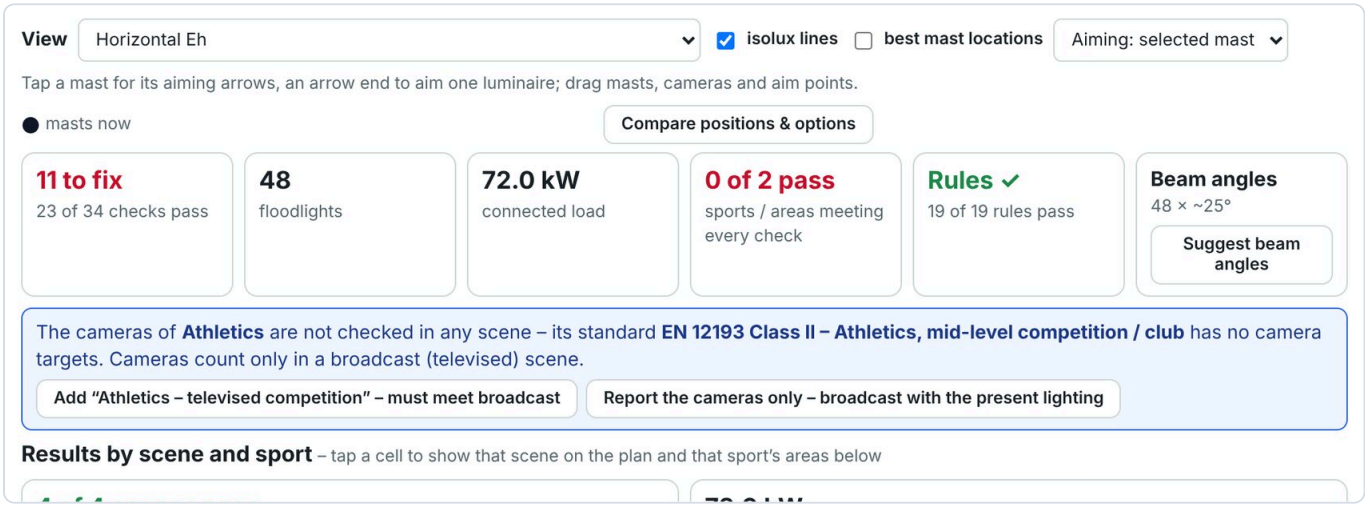
Designing and checking

Complete the design automatically – with the masts and catwalk fixed, it sized the floodlights on each, aimed and optimised them for every scene, and set each scene's level. 88 seconds.

48	72.0 kW	40 / 28.5 m	4 of 4
floodlights – 8 on the corner masts, 40 on the catwalk	connected load, all on	corner masts / catwalk	must and should scenes pass

SCENE	REQUIREMENT	LEVEL SET	KW	RESULT
Athletics – competition	must pass	65 %	46.8	pass
Athletics – training	should pass	35 %	25.2	pass
Football – match	must pass	50 %	36.0	pass
Football – training	should pass	20 %	14.4	pass
Football – televised (present lighting)	report only	100 %	72.0	camera values reported
Maintenance / security	report only	–	57.0	reported

Levels were set automatically to the lowest that passes: athletics competition 100 → 65 %, training 65 → 35 %, football match 85 → 50 %, training 35 → 20 %. All on, the playing area averages 433 lx and the track 314 lx.



Results. Light Calc notices that athletics' cameras are in no scene and offers a televised athletics scene or a report of the cameras. The top cards check All on against each sport's standard; with scenes, the scene table is what decides.

LESSON – THE BROADCAST TRAP

With the televised football scene set to **must** meet the international broadcast class (1800 lx to the cameras), the same run grew to 628 floodlights and 942 kW – and still missed by 20 lx. Reporting the cameras from the club design gave the broadcaster their values for 72 kW.

LESSON – THE SECURITY SCENE

The suggested security scene draws 57 kW – 79 % of the total. Set it to one group at 10–20 % before the obtrusive-light check: it is also the after-curfew scene.

CHAPTER 25

The neighbours, the stands and TV

Obtrusive light – zone E3, curfew 22:30

WINDOW	EV BEFORE	EV AFTER	I MAX	RESULT
No. 41 Park Road	0.4 lx	0.4 lx	13,792 cd	I over 10,000 / 1,000 cd
Flats, Park Road (5 floors)	1.5 lx	1.5 lx	128,267 cd	I over
No. 7 Lake Road	12.7 lx	10.0 lx	241,117 cd	Ev and I over
School, Lake Road	7.4 lx	5.8 lx	129,756 cd	Ev after and I over
Upward light ratio		0.4 %	limit 15 %	pass
Drivers' TI, Park Road		0.2 %	limit 15 %	pass

The light on the Park Road windows is low, but they look straight into catwalk floodlights aimed across the field. Lake Road and the school, across the field from the south catwalk, get both spill and intensity – and the after-curfew security scene keeps them over.

WHAT TO DO

(1) Make the security scene one group at 10–20 %, or all off after curfew. (2) Fit rear visors to the south-catwalk floodlights facing Lake Road and load the visor photometry. (3) Aim those floodlights 2–3° lower. Re-check: the windows update at once.

Stands – all on

SECTION	EH SEATS	FACES	SHARE	GR
North, roofed	6 lx	32 lx	10 %	36
West, grass bank	7 lx	15 lx	4 %	24
South, roofed	4 lx	4 lx	1 %	31
East, grass bank	9 lx	18 lx	5 %	18

Faces towards the main camera vs the pitch's 334 lx. Below the 25 % crowd-shot guidance everywhere – the design lights the field, not the crowd. A few floodlights aimed at the stands, or dedicated spectator lighting, would fix it. No mast floodlight loses 5 % of its beam to a roof.

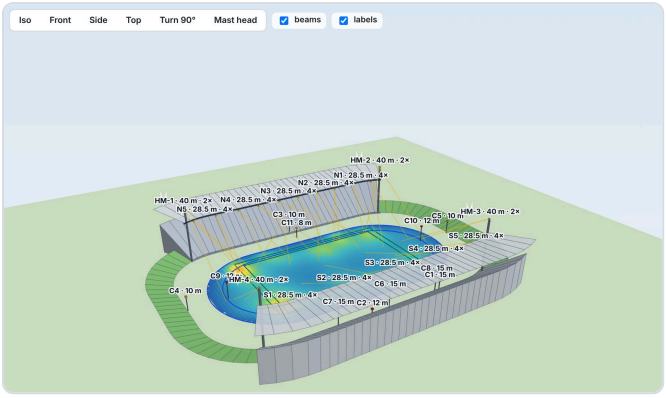
Colour and flicker – UHD

CHECK	VALUE	TARGET	
Ra	90	≥ 90	✓
TLCI	91	≥ 90	✓
CCT	5700 K	5000–6000	✓
Frame rate	1000 fps	≥ 600	✓
Flicker on the pitch	1.01 %	≤ 1 %	✗

2.5 % per floodlight, spread over three phases, leaves 1.01 % on the pitch – just over. On one phase it would be the full 2.5 %. A driver with lower flicker (≤ 2.4 %), or better phase balance on the catwalk, passes.

CHAPTER 26

Pictures, report and lessons

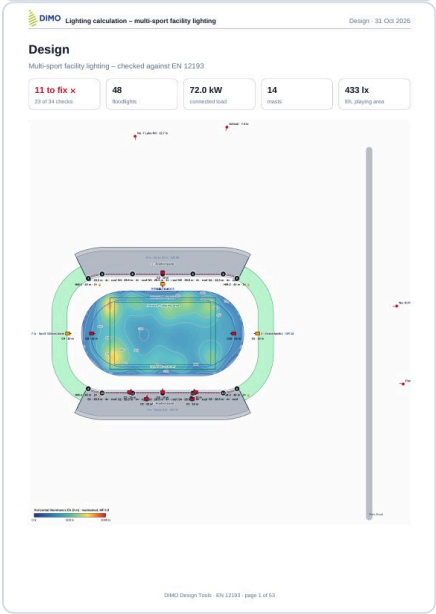


3D – masts, catwalk runs and beams with labels.



Realistic, day – for the client.

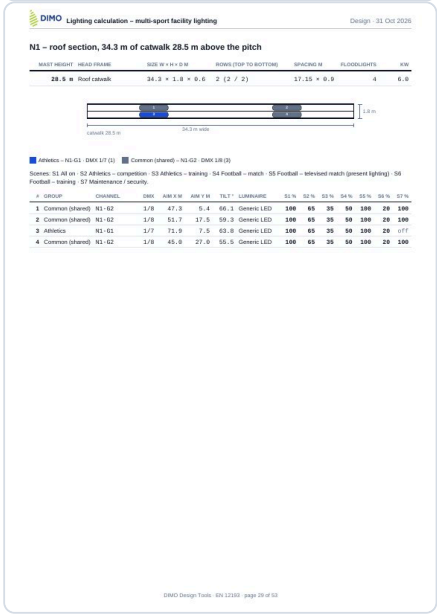
The report – 53 pages



Summary and plan



Obtrusive light



Roof section N1 – its floodlights along the catwalk

Lessons

1. **Set scenes first.** The automatic design then designs every scene and finds each one's lowest level – here saving up to 80 % in training.
2. **Decide what TV requires before designing.** Must-pass broadcast: 942 kW. Reported camera values: 72 kW.
3. **Check the neighbours early.** The security scene and two catwalk directions decide planning approval – both cheap to fix now, costly after the floodlights are bought.
4. **Light the stands on purpose** if crowd shots matter – spill alone gives 1–10 % of the pitch.
5. **Ask the maker for flicker and TLCI** early – a 0.01 % miss on flicker is a driver choice.

APPENDIX A

Limits and targets used

Obtrusive light – EN 12464-2 table 2 / CIE 150

ZONE	EV ON WINDOWS (BEFORE / AFTER CURFEW)	I PER FLOODLIGHT	ULR	TI (LAV)
E1 natural	2 / 0 lx	2,500 / 0 cd	0 %	15 % (0.1 cd/m²)
E2 rural	5 / 1 lx	7,500 / 500 cd	5 %	15 % (1)
E3 suburban	10 / 2 lx	10,000 / 1,000 cd	15 %	15 % (2)
E4 city centre	25 / 5 lx	25,000 / 2,500 cd	25 %	20 % (5)

The planning authority may set its own figures – type them under *Limits from the planning authority*; the report then says so.

Guidance values (editable, not standard requirements)

- Crowd shots: faces to the main camera 25 % of the pitch's camera illuminance.
- Colour and flicker: as in chapter 19.
- Run-off: 60 % of the playing-area value as a design allowance where the standard sets none.

Mast rules – football (defaults)

At least 4 masts; none behind the goals within the goal width ± the opening; at least 5 m beyond the run-off; aiming within 70° of vertical; masts at least the minimum distance apart; layout symmetrical for play in both directions.

Appendix B – Glossary

TERM	MEANING
Eh	Horizontal illuminance on the playing surface (lx)
Ecam, Ev	Illuminance on a plane facing a camera / a window (lx)
U1, U2	Uniformity: Emin / Emax and Emin / Eave
MF	Maintenance factor – maintained = initial × MF
GR	Glare rating, CIE 112 (lower is better; ≤ 50 for players in EN 12193)
ULR	Upward light ratio – share of the floodlights' flux above the horizontal
TI	Threshold increment – glare for drivers
CCT, Ra, R9, TLCI	Colour temperature; colour rendering index and its saturated-red sample; television lighting consistency index
Scene	A set of group levels checked against a class – match, training, televised, security
Run	A length of roof catwalk with its floodlights, designed like a mast
Must / should / report	A scene must pass to finish the design / is aimed for / is only reported
.rdf	Relux project file – exported and imported